

AWS OPERATIONS INTELLIGENCE AI

Enterprise Operations Control Center (EOCC)

Technical Architecture & Platform Design Document

Version: 1.0

1. Executive Overview

Enterprise organizations today operate hundreds of applications, thousands of cloud resources, multiple business processes, and numerous operational systems.

Organizations have already invested heavily in:

- AWS
- Splunk
- Jira
- ServiceNow
- Confluence
- GitHub
- Microsoft Teams
- Internal Runbooks
- Architecture Documentation

Despite these investments, organizations still struggle with operational visibility, ownership identification, dependency analysis, knowledge discovery, incident investigation, operational governance, and coordinated recovery.

The root problem is not lack of monitoring.

The root problem is lack of enterprise operational intelligence.

Existing systems monitor infrastructure.

Existing systems do not understand enterprises.

AWS Operations Intelligence AI is designed to solve this challenge by becoming the Enterprise Operations Control Center.

The platform continuously monitors, understands, investigates, coordinates, governs, and assists operational activities across the enterprise.

Instead of replacing existing enterprise systems, the platform acts as an intelligence and control layer above them.

2. Vision Statement

Create the operational brain of the enterprise.

The platform should continuously understand:

- Applications
- AWS Resources
- Teams
- Ownership Structures
- Business Processes
- Documentation
- Runbooks
- Historical Incidents
- Architectural Relationships
- Organizational Dependencies
- Operational Risks

and transform that understanding into actionable intelligence.

The long-term objective is to move enterprises from:

Reactive Operations

to

Intelligent Operations

and eventually

Autonomous Operations.

3. Core Design Principles

The architecture must satisfy six critical principles.

Principle 1 – Enterprise Scalability

The platform must support:

- Thousands of applications
- Thousands of AWS resources
- Millions of operational events
- Millions of logs
- Thousands of incidents
- Thousands of users

without architectural redesign.

Principle 2 – Enterprise Security

The platform will have visibility into:

- Production environments
- Incident data
- Internal documentation
- Service ownership information
- Organizational structures

Security must be treated as a first-class requirement.

Principle 3 – High Availability

The platform itself becomes mission critical.

Availability Target:

99.9%+

Principle 4 – Auditability

Every action must be traceable.

All recommendations, approvals, actions, and operational activities must be fully auditable.

Principle 5 – Explainability

AI recommendations must be explainable.

Users should understand:

- Why recommendations were generated
 - Which data sources were used
 - What evidence supports the recommendation
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Principle 6 – Human Control

The platform should assist operations.

The platform should not bypass human approval for production actions.

4. High-Level Architecture

The Enterprise Operations Control Center consists of ten primary layers.

1. Enterprise Data Sources
 2. Data Ingestion Layer
 3. Event Streaming Backbone
 4. Enterprise Knowledge Platform
 5. Enterprise Intelligence Layer
 6. Multi-Agent Intelligence Layer
 7. Operations Control Layer
 8. Security Layer
 9. Audit & Compliance Layer
 10. User Experience Layer
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5. Enterprise Data Sources

The platform integrates with existing enterprise systems.

AWS Layer

Primary operational source.

Services include:

- CloudWatch
- ECS
- Lambda
- Glue
- Step Functions
- RDS
- DynamoDB
- SQS
- SNS
- IAM Metadata
- AWS Resource Tags

Purpose:

Provide operational visibility and infrastructure context.

Splunk Layer

Purpose:

- Operational Events
 - Incident Signals
 - Historical Operational Data
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Jira Layer

Purpose:

- Incident History
 - Bug Tracking
 - Change Requests
 - Ownership Metadata
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ServiceNow Layer

Purpose:

- CMDB Information
- Change Management
- Incident Management

- Approval Workflows
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Confluence Layer

Purpose:

- Documentation
 - SOPs
 - Knowledge Articles
 - Runbooks
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GitHub Layer

Purpose:

- Repository Metadata
 - Deployment Information
 - Ownership Metadata
 - Engineering Context
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Microsoft Teams Layer

Purpose:

- Operational Communication
 - Escalation Intelligence
 - Incident Discussions
 - Team Collaboration Context
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6. Data Ingestion Layer

Dedicated microservices handle enterprise integrations.

Examples:

- AWS Connector Service
- Splunk Connector Service
- Jira Connector Service
- ServiceNow Connector Service
- Confluence Connector Service

- GitHub Connector Service
- Teams Connector Service

Each connector operates independently and can scale independently.

Hosting Platform

AWS ECS Fargate

Reasons:

- Containerized architecture
 - Serverless operations
 - Enterprise scalability
 - Operational simplicity
 - Security isolation
 - Auto scaling support
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7. Event Streaming Backbone

The platform adopts an event-driven architecture.

Direct system-to-system communication is avoided.

Benefits:

- Decoupling
 - Reliability
 - Scalability
 - Fault tolerance
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Amazon EventBridge

Acts as the Enterprise Event Bus.

Receives:

- AWS Events
- Jira Events
- ServiceNow Events
- GitHub Events
- Operational Events

Responsibilities:

- Event routing
 - Event distribution
 - Event normalization
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Amazon SQS

Acts as the enterprise messaging layer.

Responsibilities:

- Reliable delivery
- Retry handling
- Dead-letter processing
- Load buffering

Flow:

Event Source → EventBridge → SQS → Processing Services

8. Enterprise Knowledge Platform

This is the core intellectual asset of the platform.

The objective is to create a continuously updated Enterprise Knowledge Graph.

Enterprise Knowledge Graph

Stores relationships between:

Business Process ↓ Application ↓ AWS Service ↓ Owner Team ↓ Manager ↓ Runbook ↓ Documentation
↓ Historical Incidents ↓ Dependencies

The graph becomes the operational memory of the enterprise.

Database Strategy

Phase 1

AWS RDS PostgreSQL

Reasons:

- Enterprise reliability
 - Simplicity
 - Lower operational overhead
 - Strong transactional support
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Phase 2

Neo4j

Introduced when graph complexity grows significantly.

Supports advanced relationship traversal and dependency analysis.

9. Enterprise Intelligence Layer

Provides enterprise-wide search and knowledge retrieval.

Vector Search

Technology:

PostgreSQL + pgvector

Stores:

- Runbooks
- Confluence Content
- Jira Records
- Architecture Documents
- Incident Reports

Purpose:

Semantic enterprise search.

Enterprise Search

Technology:

OpenSearch

Provides:

- Operational Search
- Knowledge Search
- Incident Search
- Enterprise Search

10. Multi-Agent Intelligence Layer

The heart of the platform.

Built using:

LangGraph

Reasons:

- Stateful workflows
- Multi-agent collaboration
- Human approval workflows
- Enterprise orchestration

Ownership Intelligence Agent

Responsibilities:

- Ownership discovery
- Escalation path generation
- Team mapping

Questions Answered:

- Who owns this service?
- Which team supports this application?
- Who should be notified?

Dependency Intelligence Agent

Responsibilities:

- Relationship analysis
- Impact analysis
- Dependency discovery

Questions Answered:

- What is impacted?
 - Who is impacted?
 - Which systems depend on this service?
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Incident Investigation Agent

Responsibilities:

- Evidence gathering
- Correlation analysis
- Timeline generation
- Incident investigation

Outputs:

- Investigation reports
 - Incident timelines
 - Impact summaries
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Knowledge Agent

Responsibilities:

- Enterprise search
 - Knowledge synthesis
 - Runbook retrieval
 - Documentation analysis
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Architecture Agent

Responsibilities:

- Application architecture understanding
 - Service relationship discovery
 - Data flow analysis
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Governance Agent

Responsibilities:

- Missing ownership detection
 - Missing documentation detection
 - Missing runbook detection
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Operations Control Agent

Responsibilities:

- Operational action orchestration
 - Recovery workflow execution
 - Approval-based execution
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11. Operations Control Layer

The platform moves beyond monitoring.

The platform enables controlled operational actions.

AWS Actions

Supported actions include:

- Restart ECS Service
- Trigger Lambda
- Execute Glue Job
- Execute Step Function
- Scale ECS Tasks

Jira Actions

Supported actions include:

- Create Incident
- Update Incident
- Assign Incident
- Create Follow-Up Tasks

Teams Actions

Supported actions include:

- Team Notifications
- Escalation Messages
- Incident Updates

ServiceNow Actions

Supported actions include:

- Incident Creation
- Change Request Creation
- Workflow Updates

12. Human Approval Framework

No production action should execute automatically.

Execution Model:

AI Recommendation ↓ Human Approval ↓ Execution ↓ Audit Logging

This ensures trust, compliance, and safety.

13. Security Architecture

Security is a foundational requirement.

Identity Management

Microsoft Entra ID

Supports:

- Enterprise SSO
- Centralized Identity
- Enterprise Governance

Authentication

Protocols:

- OIDC
- SAML

Authorization

Role-Based Access Control (RBAC)

Roles:

- Viewer
- Engineer
- Team Lead
- Platform Administrator
- Executive

Secrets Management

AWS Secrets Manager

Stores:

- API Tokens
 - Credentials
 - Integration Secrets
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Encryption

At Rest:

- RDS Encryption
- S3 Encryption

In Transit:

- TLS 1.3
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14. Audit & Compliance Layer

Every activity is recorded.

Example Audit Record:

User: John Smith

Action: Restart ECS Service

Reason: High Error Rate

Approval: Platform Lead

Timestamp: UTC

The audit trail supports governance and compliance requirements.

15. User Experience Layer

Built using:

Next.js

Benefits:

- Enterprise-ready
 - TypeScript support
 - High performance
 - Excellent user experience
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16. Primary Product Modules

Enterprise Command Center

Unified operational view of the enterprise.

Displays:

- Applications
- Business Processes
- Incidents
- Ownership
- Risks

Business Process Dashboard

Monitor:

- Customer Onboarding
- Payments
- Reporting
- Order Processing

instead of infrastructure metrics.

Investigation Workspace

Displays:

- AI Investigations
- Incident Timelines
- Impact Assessments
- Recommendations

Dependency Explorer

Interactive enterprise dependency graph.

Governance Center

Displays:

- Ownership Coverage
 - Documentation Health
 - Runbook Coverage
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Operations Control Center

Provides:

- Controlled execution
 - Operational workflows
 - Incident management
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Executive Dashboard

Provides:

- Business Risk
 - Operational Health
 - Organizational Risk
 - Operational Maturity
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17. High Availability Strategy

Multi-AZ deployment across AWS.

Services:

- ECS Fargate
- RDS PostgreSQL
- Redis
- OpenSearch

Target Availability:

99.9%

18. Disaster Recovery Strategy

Backups:

- PostgreSQL
- S3
- OpenSearch

Recovery Targets:

RPO: Less than 15 Minutes

RTO: Less than 1 Hour

19. Why This Architecture Is Different

Most solutions stop at:

Logs Metrics Alerts Dashboards

The Enterprise Operations Control Center extends further.

Enterprise Systems ↓ Enterprise Knowledge ↓ Enterprise Intelligence ↓ Multi-Agent Reasoning ↓
Operational Coordination ↓ Controlled Execution ↓ Governance ↓ Business Intelligence

This transforms the platform from a monitoring system into an Enterprise Operations Control Center.

20. Long-Term Vision

The Enterprise Operations Control Center becomes the central operational operating system for the enterprise.

Instead of engineers opening multiple systems to understand:

- What happened
- Why it happened
- Who owns it
- What is impacted
- What actions should be taken

they interact with a single intelligence platform capable of continuously monitoring, understanding, coordinating, governing, and assisting enterprise operations 24 hours a day, seven days a week.

The ultimate objective is to create a trusted enterprise operational brain that improves reliability, governance, operational efficiency, and business resilience across the organization.